Results appendix — SYNTHETIC PLACEBO (no edge exists)

BTC/USDT 4h bars, 2019-09-01 to 2026-10-26. Walk-forward from 2022-01-01 with 21 quarterly refits; purge + 6-bar embargo. Test year (untouched during design): from 2025-10-27. Costs: 0.26% per round trip. Chosen model by the pre-registered rule: score. Edge validated on dev (kill switch): NO.

1. Does the model predict the trade outcome? (out-of-sample)

Model	Period	AUC target	AUC stop	Log-loss	Brier	IC exp. R
Base rate	dev	0.483	0.472	1.0041	0.6072	-0.020
Base rate	test	0.417	0.457	0.9983	0.6042	-0.104
Logistic	dev	0.539	0.523	1.0118	0.6075	-0.022
Logistic	test	0.568	0.567	0.9875	0.5935	0.014
GBM (depth 3)	dev	0.534	0.522	1.0049	0.6067	-0.018
GBM (depth 3)	test	0.559	0.553	0.9912	0.5972	-0.042

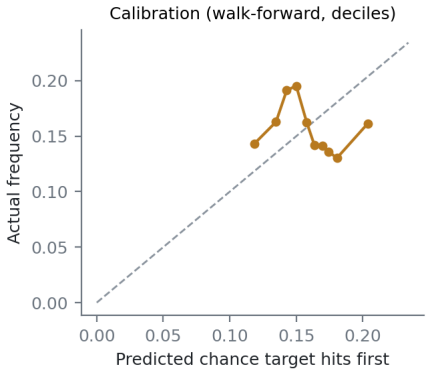
Outcome mix in the test year: target first 15.1%, stop first 34.4%, timed out 50.5%. IC = Spearman rank correlation between the model's expected result and the realised net result per bar.

2. Trading results, net of costs (one position at a time, 100% of account, no leverage)

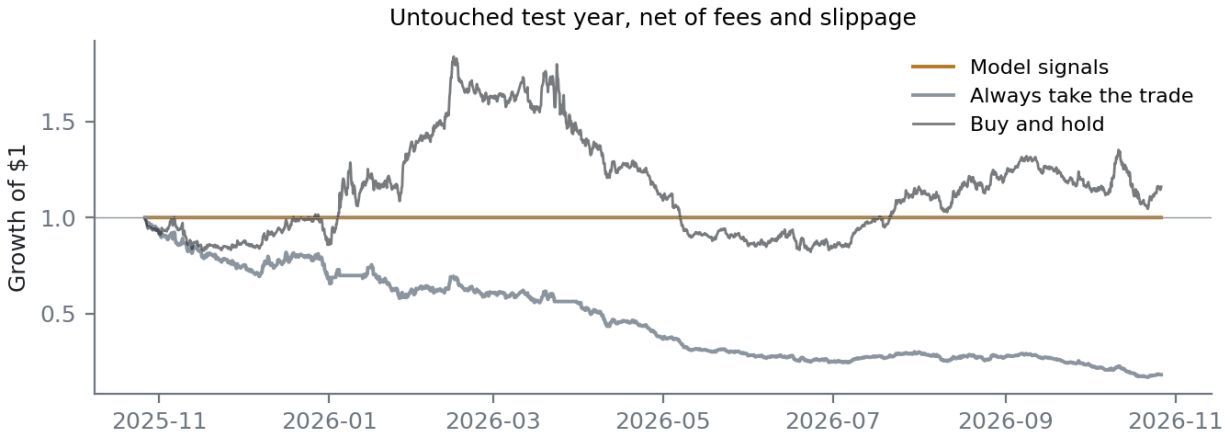
Strategy	Period	Trades	Win	Stop	Avg net R	CAGR	Sharpe	Max DD	Exposure
Model: all signals	dev	2	0.0%	0.0%	0.72	1.4%	0.88	0.0%	0.1%
Model: all signals	test	0	–	–	–	0.0%	0.00	0.0%	0.0%
Model: Medium+ only	dev	0	–	–	–	0.0%	0.00	0.0%	0.0%
Model: Medium+ only	test	0	–	–	–	0.0%	0.00	0.0%	0.0%
Always take trade	dev	1696	17.0%	35.7%	-0.13	-75.5%	-2.44	-99.6%	93.3%
Always take trade	test	436	15.4%	34.4%	-0.15	-82.0%	-3.44	-82.6%	93.1%
Buy and hold	dev	–	–	–	–	-5.6%	0.19	-64.8%	100.0%
Buy and hold	test	–	–	–	–	16.1%	0.54	-55.0%	100.0%

Configurations tried: 8. Deflated Sharpe probability (dev): 0.496 (values near 1 mean the Sharpe is unlikely to be luck from trying several variants).

Costs	Model trades	Model CAGR	Model Sharpe	Always CAGR	Always Sharpe
0.0x	0	0.0%	0.00	-43.7%	-1.01
1.0x	0	0.0%	0.00	-82.0%	-3.44
2.0x	0	0.0%	0.00	-94.2%	-5.81









3. Honest track record by confidence tier (test year)

Tier	Signal bars	Target first	Stop first	Predicted R	Realised net R
Low	0	–	–	–	–
Medium	0	–	–	–	–
High	0	–	–	–	–

Share of test-year bars with no signal: 100.0%.

4. Robustness

Check	Period	AUC target	IC	Trades	Sharpe	Always Sharpe
Barriers ×0.75	dev	0.513	-0.002	0	0.00	-3.22
Barriers ×0.75	test	0.484	-0.044	0	0.00	-5.24
Barriers ×1.25	dev	0.495	-0.016	0	0.00	-1.65
Barriers ×1.25	test	0.565	-0.067	0	0.00	-2.50
Drop 'ema_slope'	dev	0.468	-0.038			
Drop 'ema_slope'	test	0.541	-0.042			

Year	AUC target	IC exp. R	Signal coverage	Signal avg net R
2022	0.488	-0.043	0.0%	–
2023	0.431	-0.093	0.0%	–
2024	0.453	-0.027	0.0%	–
2025	0.503	-0.019	0.2%	0.50
2026	0.542	-0.050	0.0%	–
Regime: calm	0.471	-0.031		
Regime: normal	0.475	-0.055		
Regime: stormy	0.558	-0.038		

Volatility model (QLIKE, lower is better): HAR 0.411 vs EWMA 0.440 on dev; HAR 0.407 vs EWMA 0.425 on test.

Largest logistic weights (target-vs-stop, standardised): ema_slope -0.251, ret_42 +0.164, px_vs_ema +0.151, ret_24 -0.122, taker +0.104, vol_ratio +0.104.

6. Data integrity

Source: synthetic. Raw rows None, clean rows None, duplicate-timestamp rows None (corrupted duplicates kept only if consistent with neighbours), missing 4h bars None (exchange outages, all before March 2020), labels

dropped because their 24h window crossed a gap: 0. Every 4h candle was cross-checked against the independent 1h file: all but one corrupted 1h row agree within 0.2%.